

Lab 3 - Routing Protocol

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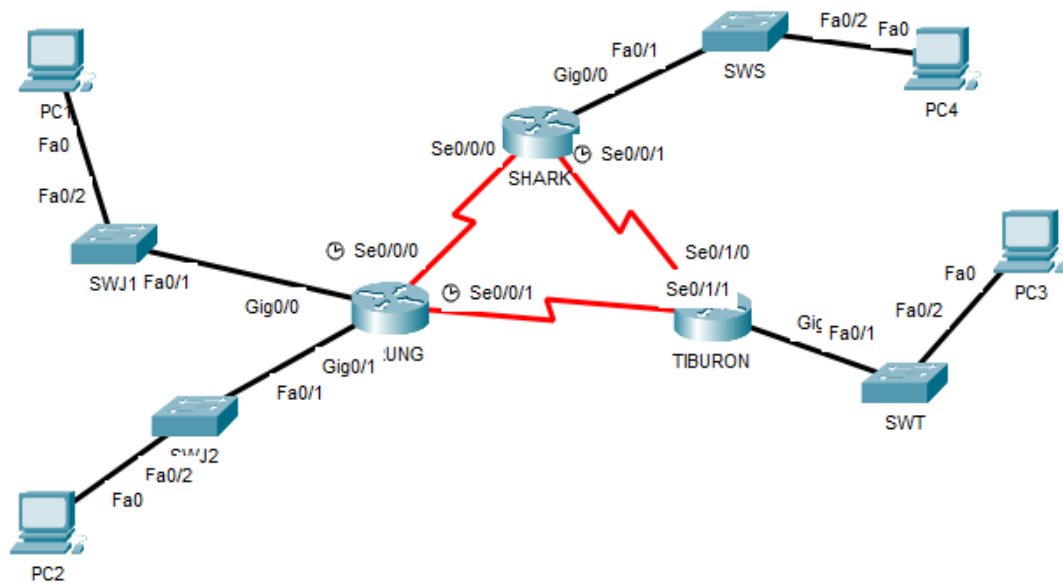
MATRIC NUMBER: A23CS0220

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - a. CLI
 - i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- JERUNG

```
JERUNG>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C    172.18.110.0/27 is directly connected, GigabitEthernet0/0
L    172.18.110.1/32 is directly connected, GigabitEthernet0/0
C    172.18.110.32/27 is directly connected, GigabitEthernet0/1
L    172.18.110.33/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
```

- TIBURON

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.128/26 is directly connected, GigabitEthernet0/0
L    172.18.110.129/32 is directly connected, GigabitEthernet0/0
C    172.18.110.196/30 is directly connected, Serial0/1/1
L    172.18.110.198/32 is directly connected, Serial0/1/1
C    172.18.110.200/30 is directly connected, Serial0/1/0
L    172.18.110.202/32 is directly connected, Serial0/1/0
```

- SHARK

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C    172.18.110.64/26 is directly connected, GigabitEthernet0/0
L    172.18.110.65/32 is directly connected, GigabitEthernet0/0
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.194/32 is directly connected, Serial0/0/0
C    172.18.110.200/30 is directly connected, Serial0/0/1
L    172.18.110.201/32 is directly connected, Serial0/0/1
```

b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

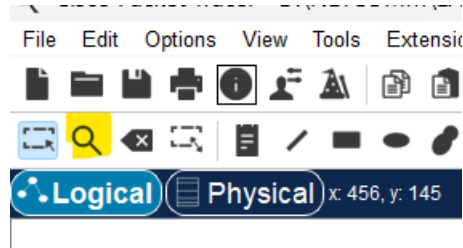


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

- JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

- TIBURON

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	

- SHARK

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

2. Try to ping PC2 from PC1, paste the results here.

```

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=7ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time=2ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

The ping result to 172.18.110.62 (PC2) was successful, with all four packets sent and received without any packet loss. This indicates that PC1 has proper connectivity with PC2. The low round-trip time (average 2ms) confirms that the network connection between these two devices is efficient. This result shows that the configuration of both PCs, their default gateways, and the routing between their respective subnets is correctly set up. The routers involved are correctly exchanging routing information, allowing communication between PC1 and PC2.

However, the ping to 172.18.110.190 (PC4) was unsuccessful, as all packets failed, with the error message "Destination host unreachable" returned by 172.18.110.1, which is PC1's default gateway (JERUNG). This indicates that JERUNG does not have a route to reach PC4's subnet (172.18.110.128/26). The issue likely arises from incomplete or incorrect RIP configurations on the routers. Either JERUNG is not learning about PC4's subnet, or TIBURON is not advertising it. Alternatively, there might be connectivity or routing table issues preventing JERUNG from reaching TIBURON and SHARK.

5. What needs to be done to ensure all PCs can ping each other successfully?

To ensure that all PCs can ping each other successfully, it is important to verify and configure proper IP addressing for each PC. This includes assigning correct IP addresses, subnet masks, and default gateways according to their respective subnets. For instance, PC1 should have an IP address of 172.18.110.30, a subnet mask of 255.255.255.224, and a default gateway of 172.18.110.1, while PC4 should have an IP address of 172.18.110.190, a subnet mask of 255.255.255.192, and a default gateway of 172.18.110.129. Any misconfiguration in addressing could lead to communication failures.

Proper routing must also be enabled on all routers (JERUNG, SHARK, and TIBURON) to ensure they can advertise and recognize all relevant subnets. Using a dynamic routing protocol like RIP is crucial. For example, JERUNG should advertise the 172.18.110.0/27 and 172.18.110.32/27 networks, SHARK should advertise 172.18.110.64/27, and TIBURON should advertise 172.18.110.128/26. Each router's RIP configuration must be carefully verified to confirm that all necessary networks are being advertised. The routing tables of each router should then be checked using the `show ip route` command to ensure routes to all PC subnets are present.

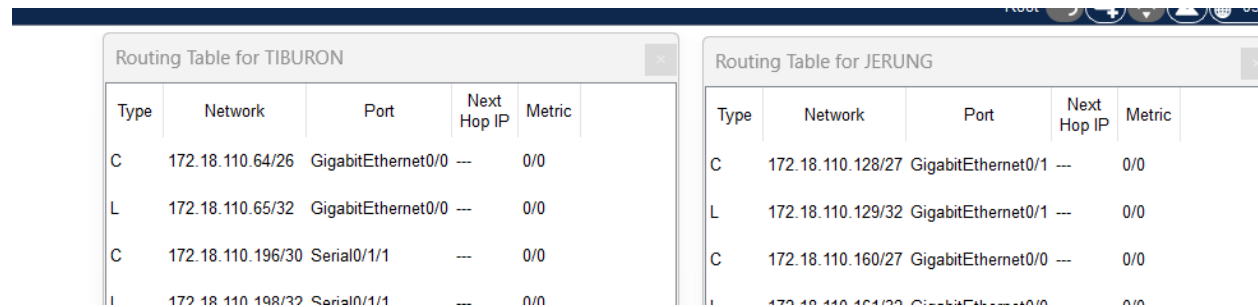
It is also essential to test the connectivity between the routers. For example, JERUNG should be able to ping TIBURON's G0/0 interface at 172.18.110.129, and SHARK should be able to communicate with both JERUNG and TIBURON. If connectivity between routers fails, interface statuses should be checked using `show ip interface brief`, and any issues with disabled or misconfigured interfaces should be resolved. Similarly, PC-to-router communication must be validated by ensuring that each PC can ping its assigned default gateway. If this fails, the PC's network configuration and the router's interface status should be verified.

Additionally, there should be no firewalls or access control lists (ACLs) blocking ICMP packets between subnets. If ACLs are in use, they should be adjusted to allow ICMP traffic. Once these steps are completed, end-to-end connectivity between PCs should be tested by pinging the IP addresses of all other PCs from each PC. For example, PC1 should ping PC4 at 172.18.110.190 and PC2 at 172.18.110.62. If any pings fail, troubleshooting should focus on routing, IP addressing, and connectivity between the devices.

Finally, once all PCs can ping each other successfully, it is advisable to save the configurations on all routers using the `write memory` command to prevent the loss of settings. By carefully addressing these steps, all PCs in the network will achieve seamless communication.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG.
This is done to show changes to the routing table as configurations are made.



Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- a. What can you say about the addresses used in the ‘network’ instructions in Figure E?

The addresses used in the network instructions in the given RIP configuration represent subnets within the IP range of 172.18.110.x. These addresses are likely subnet IDs or specific subnets that the router participates in for routing. The configuration is using RIP version 2, as indicated by the version 2 command. This version of RIP supports Classless Inter-Domain Routing (CIDR), allowing the use of subnets with specific masks rather than default classful networks. Additionally, the no auto-summary command ensures that RIP does not automatically summarize these subnets to their classful boundaries, which would otherwise aggregate them into a single larger network (e.g., 172.18.0.0).

The addresses provided 172.18.110.128, 172.18.110.160, 172.18.110.192, and 172.18.110.196 suggest that the network has been divided into smaller subnets. These subnets are being advertised and monitored by the router through RIP. The increment patterns in the addresses, such as 128 to 159 or 160 to 191, indicate the possibility of a /27 or similar subnet mask being used. These specific subnets allow the router to dynamically share and update routing information for these networks with other RIP-enabled routers in the same environment. This configuration ensures efficient and flexible route management across the subnets.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

TIBURON

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1	
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1	
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	

JERUNG

Routing Table for JERUNG					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0	
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0	
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.193/32	Serial0/0/0	---	0/0	
C	172.18.110.196/30	Serial0/0/1	---	0/0	
L	172.18.110.197/32	Serial0/0/1	---	0/0	
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1	

- a. What are the changes seen in TIBURON?

The changes in the TIBURON routing table indicate that the router is now correctly learning and advertising routes. The table displays the following details:

1. Directly Connected Routes (C):

- **172.18.110.128/26:** This is directly connected via the GigabitEthernet0/0 interface, representing the subnet where PC4 resides.
- **172.18.110.129/32:** This is the IP address of the GigabitEthernet0/0 interface itself.
- **172.18.110.196/30:** This is directly connected via the Serial0/1/1 interface, representing the point-to-point link between TIBURON and SHARK.
- **172.18.110.200/30:** This is directly connected via the Serial0/1/0 interface, representing the point-to-point link between TIBURON and SHARK.
-

2. Remote Routes (R):

- **172.18.110.0/27:** Learned via RIP, accessible through Serial0/1/1 with the next-hop IP **172.18.110.197**.
- **172.18.110.32/27:** Learned via RIP, accessible through Serial0/1/1 with the next-hop IP **172.18.110.197**.
- **172.18.110.192/30:** Learned via RIP, accessible through Serial0/1/1 with the next-hop IP **172.18.110.197**.

- b. What are the Networks with type R in TIBURON and JERUNG?

- Tiburon

- Network 172.18.110.0/27
- Network 172.18.110.32/27
- Network 172.18.110.192/30

- Jerung

- Network 172.18.110.128/26
- Network 172.18.110.200/30

- c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

- d. Ping PC4 from PC1. Was it successful?

No

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- e. Explain the reasons for your answer.

For 172.18.110.126, the "Destination host unreachable" message means that another device on the network (probably the router at 172.18.110.1) couldn't find that IP address. This happens when the device isn't connected to the network, its IP is wrong, or there's a problem with the network's routing.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```

SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.196
SHARK(config-router)#no auto-summary
SHARK(config-router)#

```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
----	-------------

1	<pre> C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=12ms TTL=126 Reply from 172.18.110.30: bytes=32 time=5ms TTL=126 Reply from 172.18.110.30: bytes=32 time=3ms TTL=126 Reply from 172.18.110.30: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 5ms </pre>
2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=6ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 6ms, Average = 2ms </pre>
3	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=9ms TTL=126 Reply from 172.18.110.190: bytes=32 time=3ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 9ms, Average = 3ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
No. Both of PCs cannot ping PC3

```

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

SHARK

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0

R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

Referring to the routing table, explain your findings.

The routing tables for TIBURON and SHARK show how these routers communicate with the rest of the network. In TIBURON's table, the network 172.18.110.196/30 is directly connected to the router through its Serial0/1/1 interface. This is why it's marked with a C for "connected." Other networks, like 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.64/26, are learned through RIP, as indicated by the R label. These routes use the next-hop IPs 172.18.110.197

and 172.18.110.201 to reach their destinations, and the metric of 120 means they are one hop away.

Similarly, SHARK's routing table includes both directly connected and RIP-learned routes. Networks like 172.18.110.64/26 and 172.18.110.192/30 are directly connected to SHARK through its GigabitEthernet0/0 and Serial0/0/0 interfaces. These directly connected networks are marked with a C. SHARK also shows local routes, such as 172.18.110.65/32 and 172.18.110.194/32, which represent the IP addresses assigned to its interfaces. The RIP-learned routes, 172.18.110.0/27 and 172.18.110.32/27 are advertised by another router with the next-hop IP 172.18.110.193, and they are also one hop away.

In short, both routers use RIP to share information about the networks they're connected to. This allows them to "learn" about other networks beyond their direct connections. If a new network, like 192.168.1.0/24, is added, it should also be advertised via RIP so all routers in the network can recognize and use it.

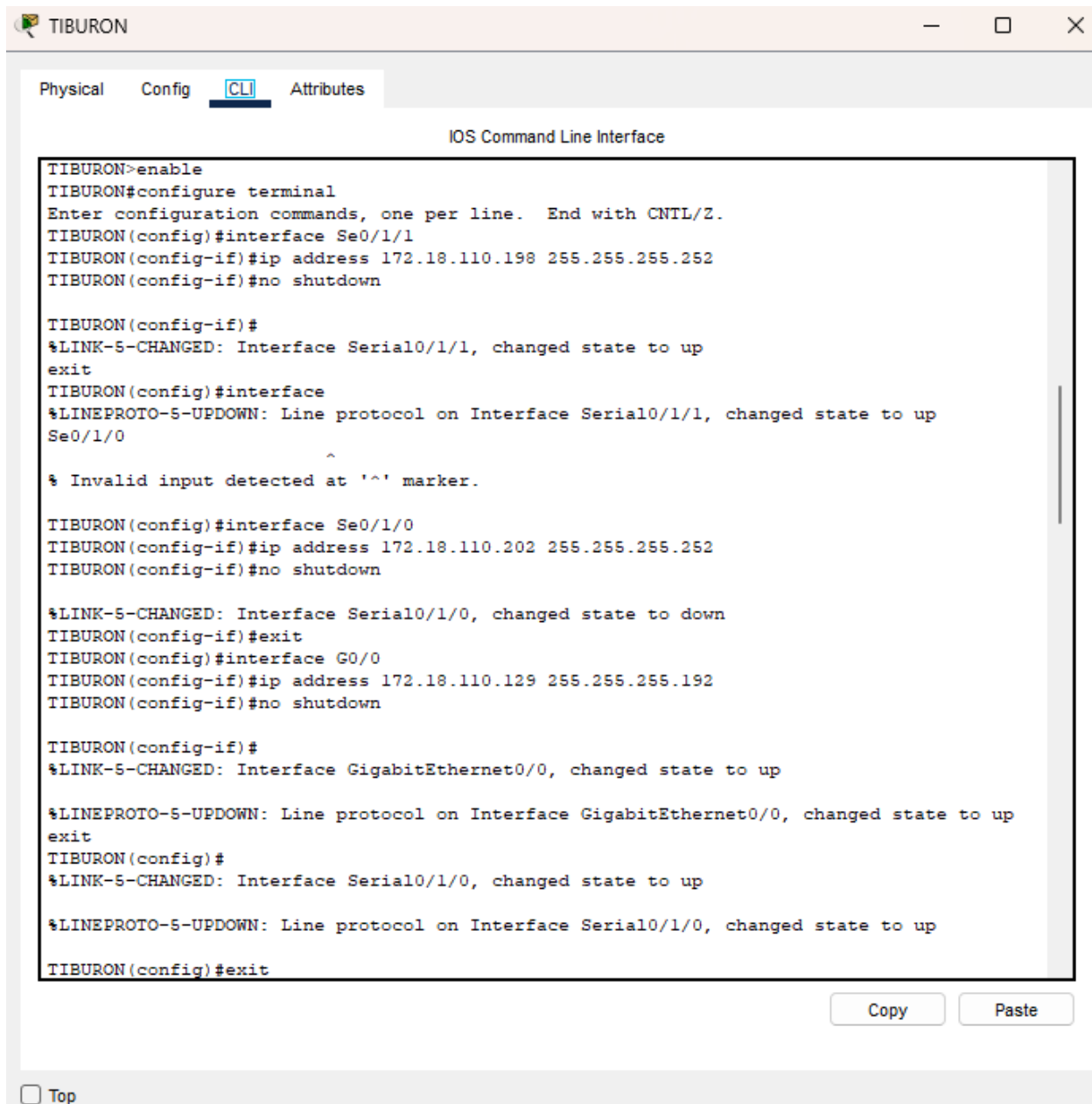
- e. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

To ensure all PCs can communicate with each other successfully, the first step is to update the RIP configuration on all routers to include any new networks, such as 192.168.1.0/24 (the network for PC3). This ensures the routers share information about the new network. On each router (TIBURON, SHARK, and JERUNG), I would enter commands to advertise the new network. After that, it's important to check the routing tables using the `show ip route` command to confirm that the new network appears on all routers.

Next, I need to make sure all PCs are configured correctly with the proper IP addresses, subnet masks, and default gateways. For example, PC3 should have an IP address of 192.168.1.2, a subnet mask of 255.255.255.0, and a default gateway of 192.168.1.1 (TIBURON's G0/0 interface). Once everything is set up, I can test connectivity by using the `ping` command from each PC to ensure they can reach the others. If any pings fail, it may be necessary to troubleshoot the network, such as

checking for incorrect configurations or any settings that might block traffic.
Following these steps will ensure full connectivity across the network.

- f. Show your configurations in TIBURON to ensure end-to-end connectivity.



The screenshot shows the TIBURON web interface with the 'CLI' tab selected. The window title is 'TIBURON'. Below the tabs, the text 'IOS Command Line Interface' is displayed. The main content area shows a terminal session with the following commands and output:

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface Se0/1/1
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#no shutdown

TIBURON(config-if)#
%LINK-S-CHANGED: Interface Serial0/1/1, changed state to up
exit
TIBURON(config)#interface
%LINEPROTO-S-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
Se0/1/0
^
% Invalid input detected at '^' marker.

TIBURON(config)#interface Se0/1/0
TIBURON(config-if)#ip address 172.18.110.202 255.255.255.252
TIBURON(config-if)#no shutdown

%LINK-S-CHANGED: Interface Serial0/1/0, changed state to down
TIBURON(config-if)#exit
TIBURON(config)#interface G0/0
TIBURON(config-if)#ip address 172.18.110.129 255.255.255.192
TIBURON(config-if)#no shutdown

TIBURON(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
TIBURON(config)#
%LINK-S-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
TIBURON(config)#exit
```

At the bottom right of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' link with a small square icon next to it.

- g. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre> C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), </pre>
2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), </pre>
4	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), </pre>

REFLECTION

I think I've gained a good understanding of how to use Packet Tracer and the different functions it offers. While working on the configurations, I realized how important it is to be precise, especially when setting up IP addresses on routers and devices. Even a small mistake, like a typo, can cause the entire network to not work, so I've learned to be more careful and double-check my work.

I also got to see how networks are structured and how devices communicate with each other in Packet Tracer. It gave me a clearer idea of how everything works together, from the routers to the end devices. This experience helped me develop better problem-solving skills because I had to figure out where things went wrong when the network didn't function as expected. Overall, it's been a really hands-on way to learn about networking, and I feel much more confident analysing and troubleshooting networks now.

.